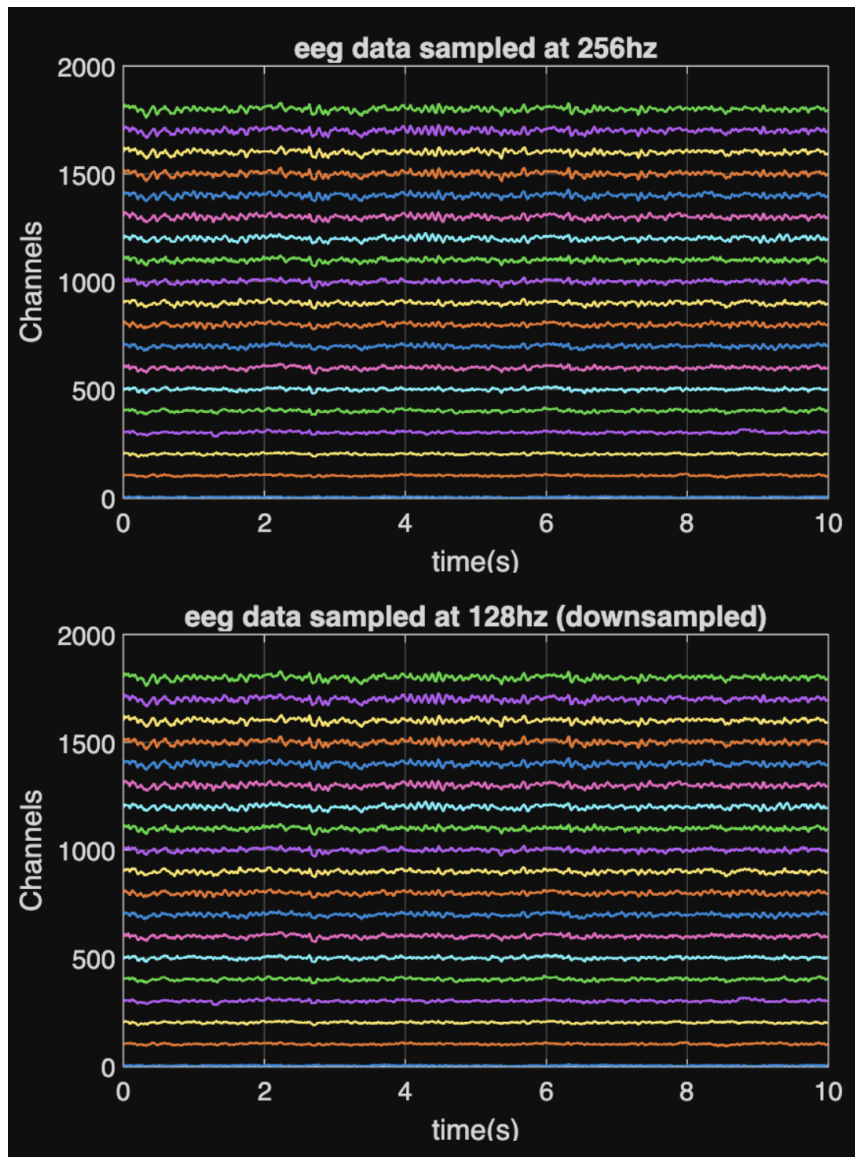


TASK 0:

An EEG file (.mat format) is attached with this mail for performing the following tasks. The file has 19 channel EEG recordings of a participant which is of duration 10 seconds at a sampling frequency 256 Hz (Dimension: 19 X 2560) . The data is already preprocessed (i.e.) band-pass filtered, eye and muscle artefacts removed.

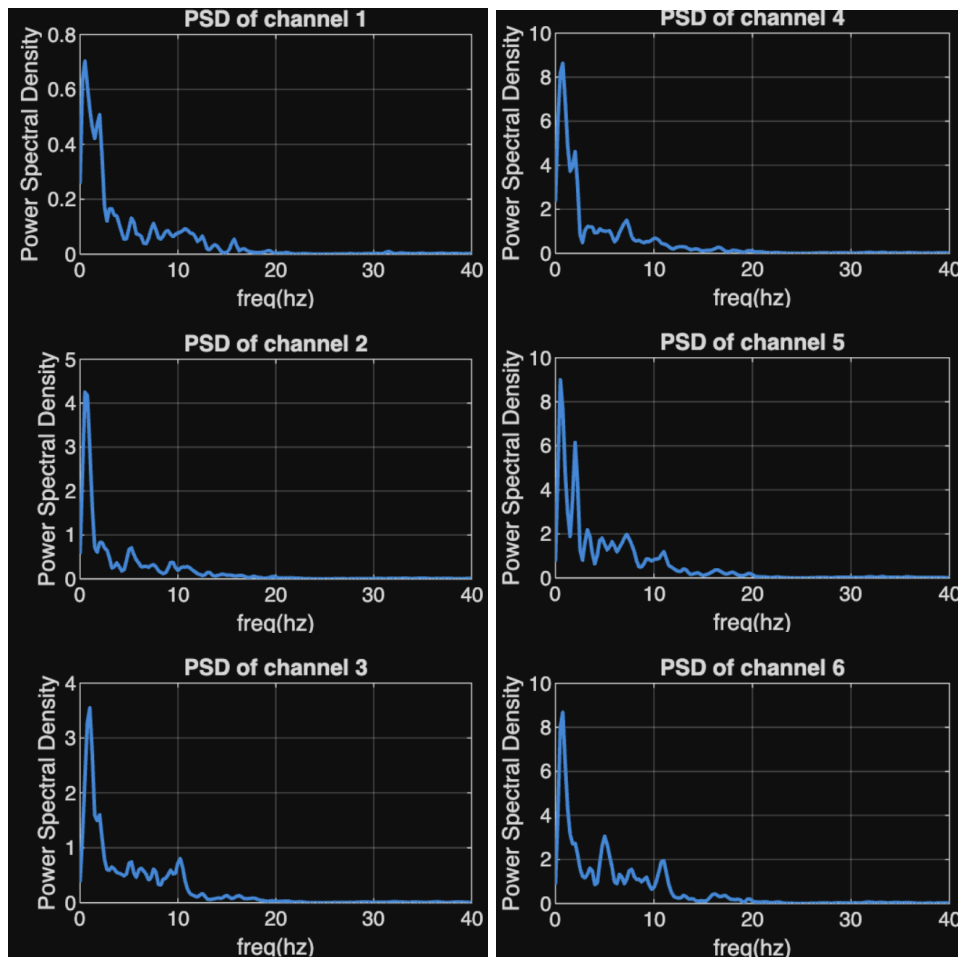
Plot the EEG signals before and after downsampling.

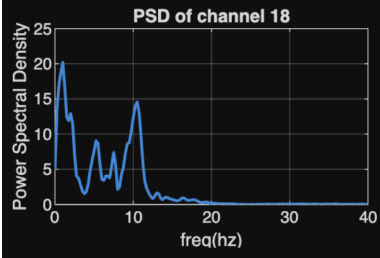
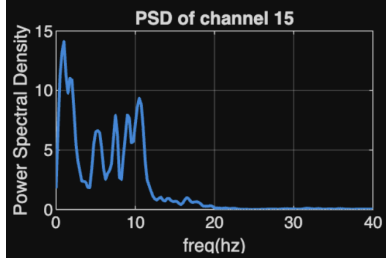
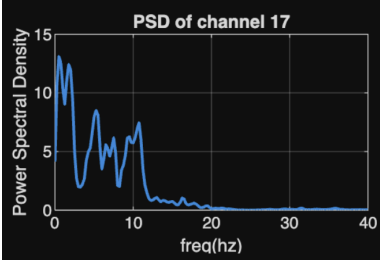
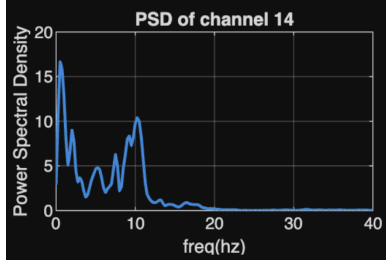
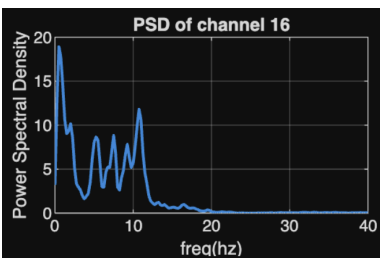
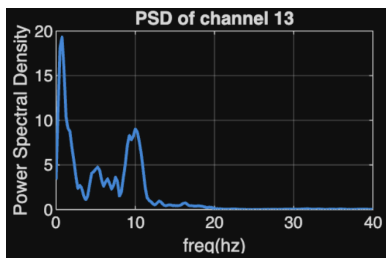
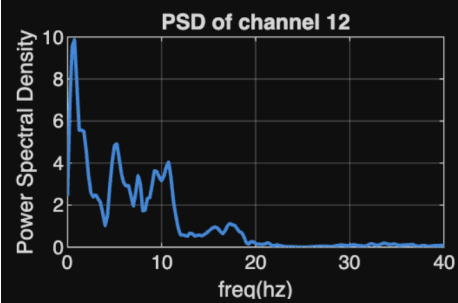
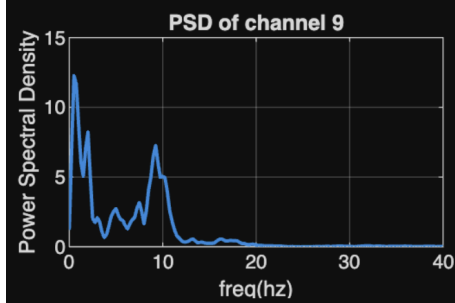
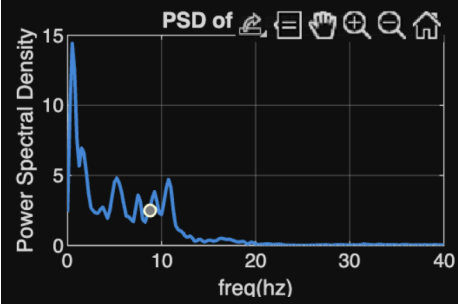
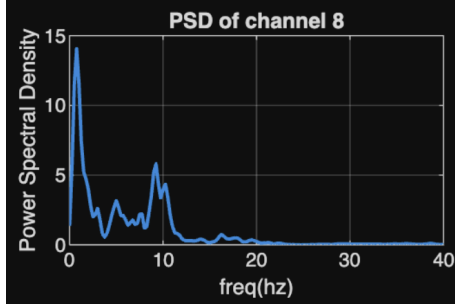
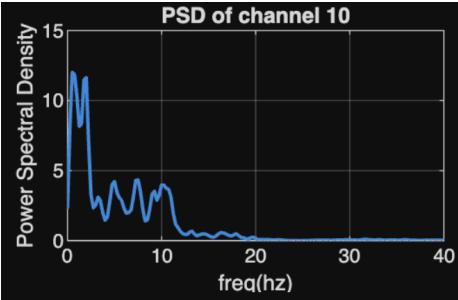
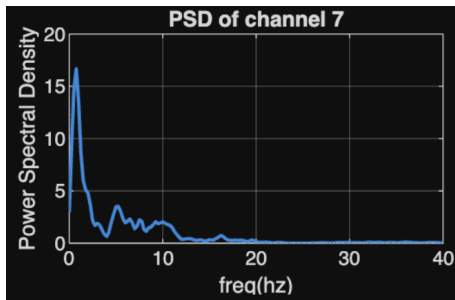


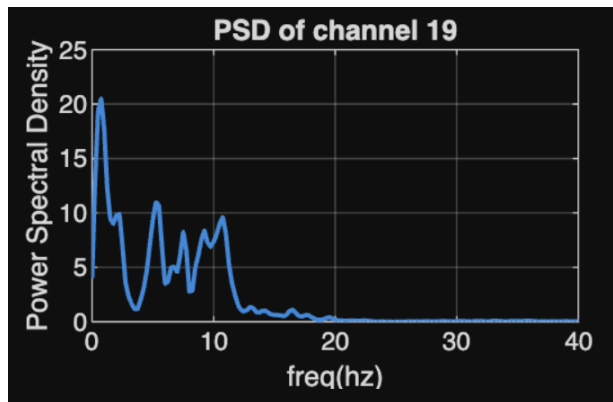
TASK 1:

Power Spectral Density (PSD) Analysis

- Compute the Power Spectral Density (PSD) for each EEG channel in the provided preprocessed signal.
- Use appropriate techniques (e.g., Welch's method) to calculate the PSD and plot corresponding figures for each channel as subplots.



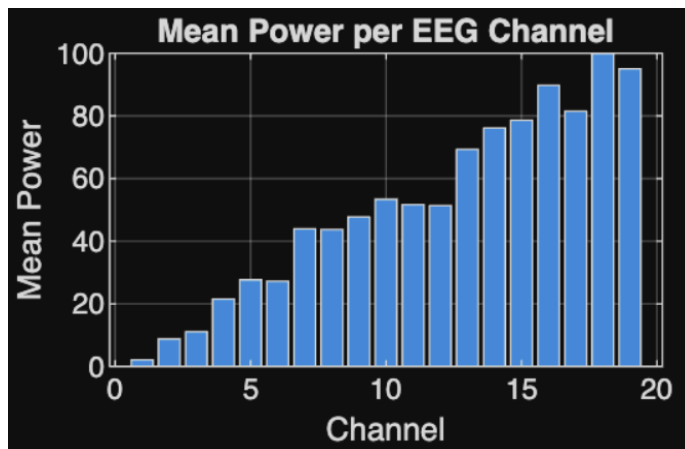




TASK 2:

Mean Power Calculation and Visualization

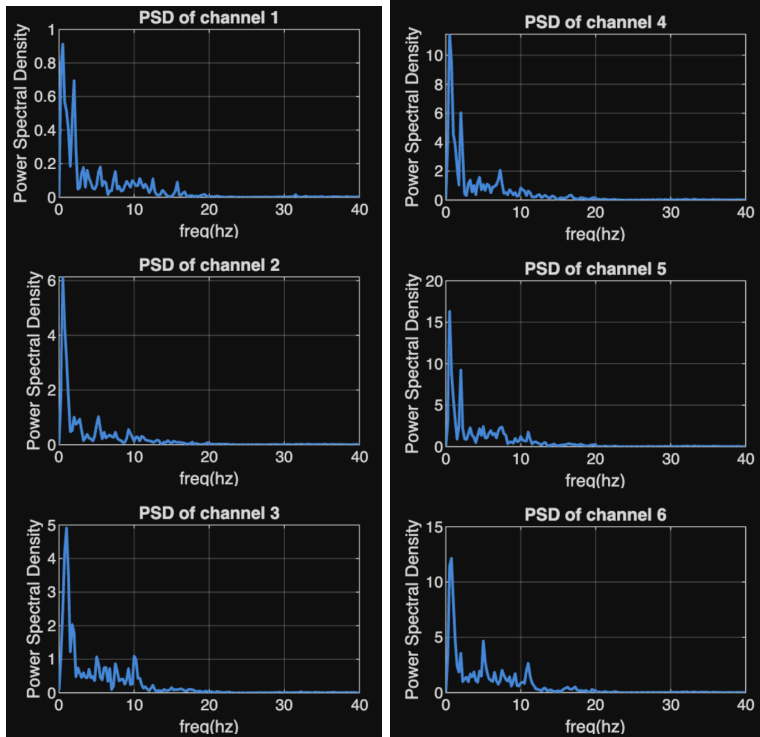
- Calculate the mean power of each channel across the signal.
- Plot the mean power values for all channels in a clear and labeled graph (line/bar/ of your choice).

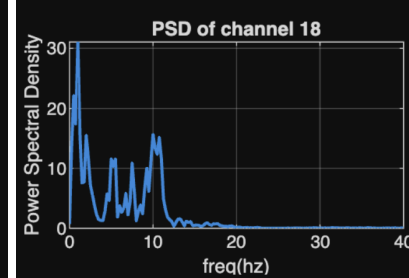
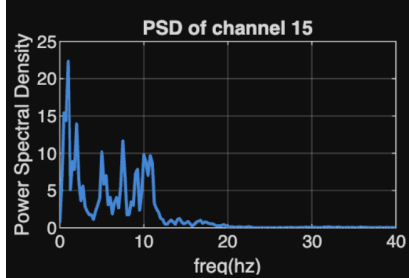
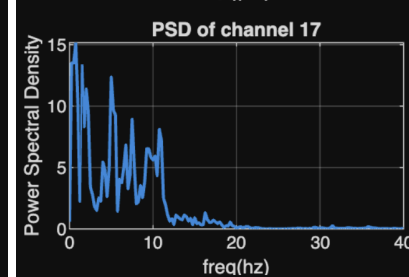
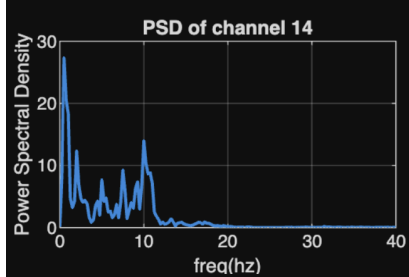
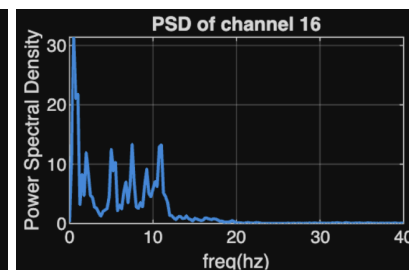
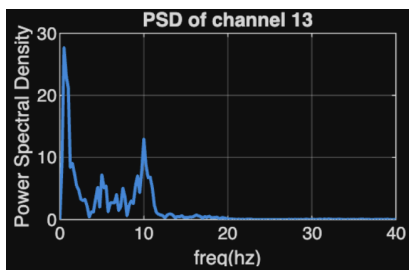
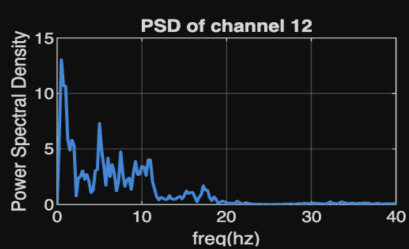
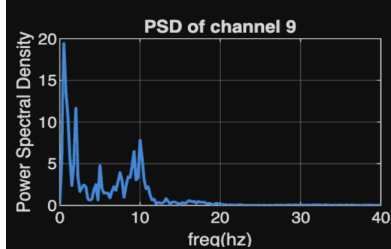
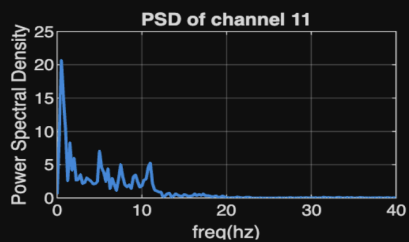
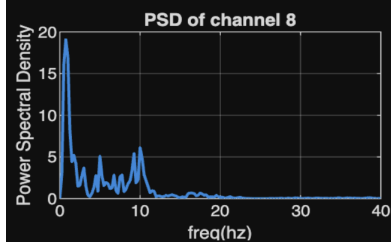
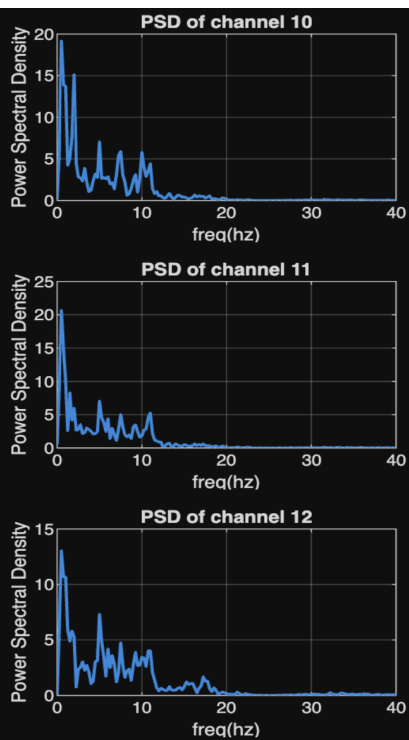
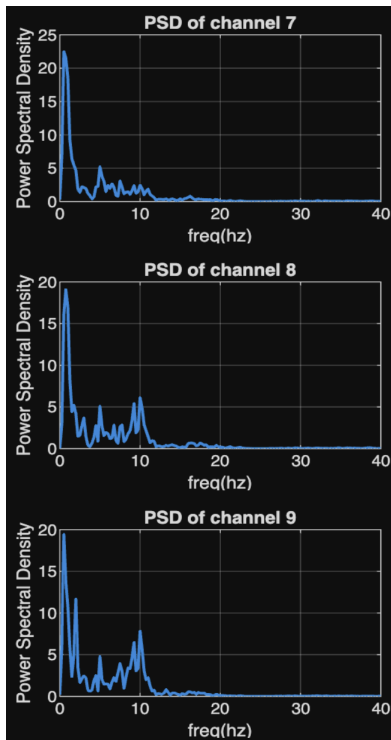


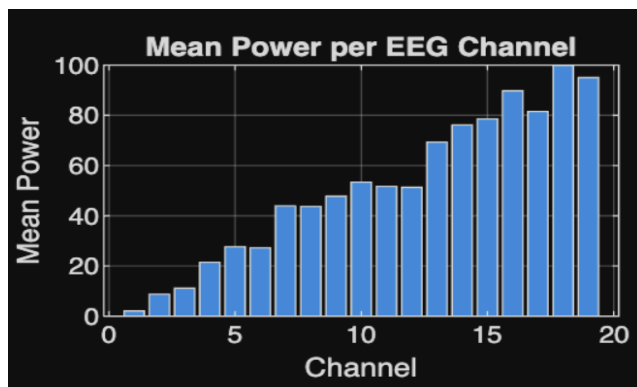
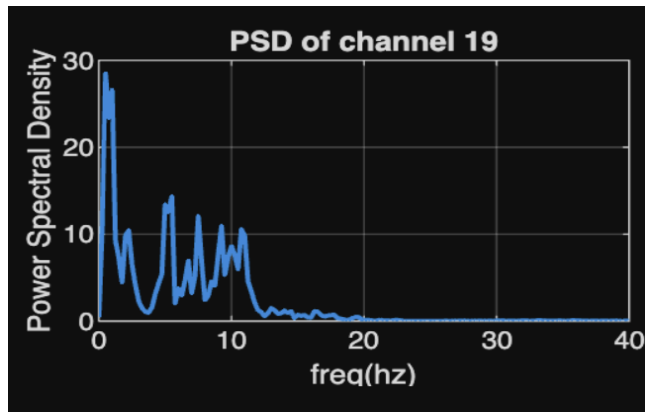
TASK 3:

Downsampling and Reanalysis

- Downsample the EEG signal to 128 Hz while preserving signal integrity.
- Repeat Tasks 1 and 2 on the downsampled signal.







KEY OBSERVATIONS AND INFERENCES

TASK 0:

FIG1:

EEG data sampled at 256hz. All 19 channels are plotted with vertical offsets so they do not overlap.

The total time window is 10 seconds.

At 256 Hz sampling, each channel contains 2560 samples, so the signal appears dense.

FIG2:

EEG data sampled at 128 Hz (downsampled by a factor of 2). The matrix size reduces from 19x2560 to 19x1280, since we keep every 2nd sample. This reduces the density of the signal but the shape looks similar. Downsampling to 128 Hz does not cause information loss because eeg useful frequency content is ≤ 40 z. By Nyquist criterion:

$$f_s \geq 2f_m$$

Our downsampled rate is 128hz(>80hz).

Therefore downsampling to 128hz preserves all important eeg information.

TASK 1:

FIG 3–21:

What is psd?

The power spectral density of a signal is a measure of how the signal's average power is distributed across its frequency components.

It provides a frequency-domain view of the signal, showing which frequencies contribute the most power.

Isn't fourier transform enough why psd?

The fourier transform shows which frequencies are present in a signal. it does not reliably show how much power each frequency carries. Its amplitude depends on window length, noise, and scaling, making it unstable for real eeg data.

psd provides a more meaningful representation by showing how the avg power of the signal is distributed across frequencies. psd smooths fluctuations, reduces noise effects, and gives a consistent estimate of power. because eeg is noisy, psd is preferred for identifying meaningful brainwave bands such as delta, theta, and alpha.

The welch psd shows how power is distributed across frequencies. Most channels show strong activity in the 0–15 Hz range.

So delta theta and alpha waves can be seen strongly which imply activities ,deep sleep, light sleep and a relaxed state.

formula for psd:

$$S_x(f) = \frac{1}{N \cdot f_s} |X(k)|^2$$

TASK 2:

Mean power formula:

$$\frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$$

Each bar represents the average signal energy of a channel over 10 seconds. Channels with higher mean power show stronger activity.

psd vs mean power:

psd shows power distribution across frequency, while mean power shows total energy over time.

TASK 3:

FIG 23–41:

PsD shape has remained almost identical in shape. The only difference is slightly lower frequency resolution due to fewer samples. But interpretation remains unchanged.

FIG 42:

Similar to observations in psd, the shape remains same. Mean power is nearly unchanged.

My inference: use downsampled signals for ml processing to work which reduces computational load computational. No info is lost in the downsampling of the signal (condition: satisfy sampling theorem). For eeg, meaningful activity is only upto 40hz, so any sampling freq ≥ 80 hz is suitable.